

Introduction

Partial dependence plots (PDPs), introduced by Friedman in 2001, show the average model prediction as a feature varies while holding other features at their observed values. They provide a visual answer to “how does the model’s prediction change with x_j ?” — exposing non-linear effects and interactions in black-box models that variable-importance scores alone cannot reveal. Combined with variable importance (which features matter) and SHAP values (per-prediction explanations), PDPs are

```
xlab = "x1", ylab = "PD(x1)",  
main = "Partial dependence of x1")
```

Output & Results

`pdp::partial()` returns a data frame with the feature value and the partial-dependence prediction. Plotting it as a curve shows how the average prediction varies

- Imaging and sequence models intro
- tidymodels pipelines
- FDR, knockoffs, replication crisis